Big Data and Machine Learning for Applied Economics

Week 02: Linear Regression as a Prediction Machine — and How to Evaluate a Prediction

Eduard F. Martinez Gonzalez

Department of Economics, Universidad Icesi

October 12, 2026

Previously...

- **Y, not β:** predicting and explaining are different questions; a significant coefficient is not a prediction.
- The goal is the **generalization risk** (under squared loss, the **MSE**); the best predictor is the conditional mean $\mathbb{E}[Y \mid X = x]$; σ^2 is the ceiling.
- **Golden rules:** #1 the training error is optimistic; #2 the test set is used at the end, exactly once.
- $\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma^2$: complexity lowers the bias and raises the variance; the optimum is interior (the U curve).
- **We saw it with data:** 50 listings, three polynomials; degree 12 fit best in sample and predicted worst out of sample. Today the course's problem begins.

Today

The regression you already know, as a **prediction machine**: what carries over from OLS, what stops mattering, what is new; k-NN as a first contrast; and how to **evaluate** a prediction: metrics, benchmark, figures, where to look at the errors.

Today's question: how much of a station's vote can its surroundings predict?

The course's case: the 2022 presidential runoff, Petro vs. Hernández, station by station

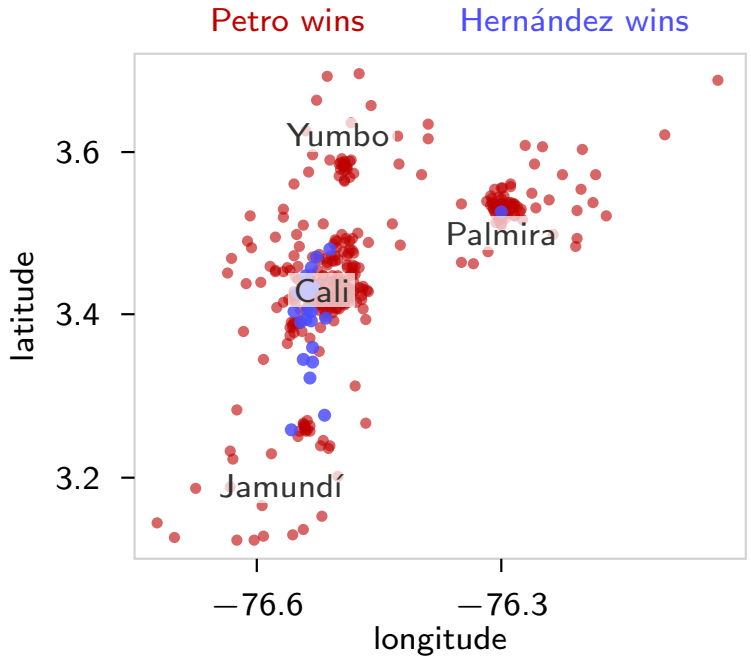
- **The unit:** the polling station (a school, a community hall): 12,000 in the country with coordinates and census data; 1,060 in Valle; 343 in Cali, Palmira, Yumbo, and Jamundí.
- **Today's y, and sessions 3 and 5's:** `voto_petro`, Petro's votes over the two candidates' combined votes, in points. From 20% (Santander) to 86% (Chocó); in the metro area, from 15% (Pance) to 94% (La Meseta, Jamundí).
- **Session 4's y:** `gana_petro`: did Petro get more votes than Hernández? That is what winning means, not clearing 50%.
- **The x's:** what the 2018 census says about the station's **surroundings**. No polls, no campaign data: structure only.

How we answer it today, and throughout the course

A campaign wants to know which stations are "theirs" before spending; an analyst, how much of the vote is the territory's *structure*. Today: the theory of regression as a prediction machine, with these stations as the example, and then the same exercise in R. Every week we return to the case with one more tool; a **leaderboard** keeps score.

The question, on the map

Each dot is a polling station; the color, who won it (more votes than the rival)



- The vote has an unmistakable geography: eastern Cali and Jamundí, nearly unanimous; southern Cali, the other shore. And none of this uses polls: it is just the outcome.
- The campaign's question: how much of this picture could be anticipated *with the census alone*? To answer it, first we need to know what we know about each dot.

What we know about each station

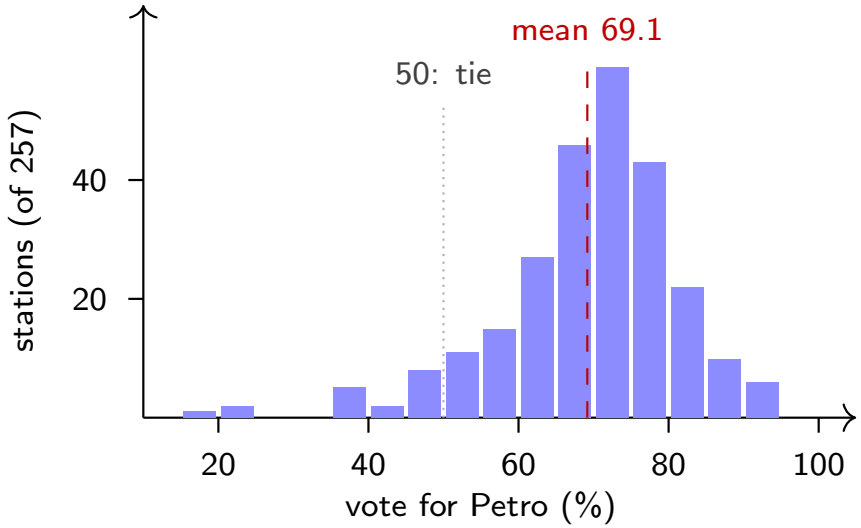
The census blocks assigned to each station, summarized in four groups (plus location)

Group	**Variables (as % of the surroundings)**	**In Cali, Palmira, Yumbo, and Jamundí**
Socioeconomic	low- (1–2), middle- (3–4), and high-stratum (5–6) housing; people with higher education	higher education: 0 to 58%; high stratum: 0 to 92%
Infrastructure and size	utilities per household (water, sewerage, power, gas, internet: 0 to 5); population; registered voters	registered: from 44 to 22.000
Demographics	six age groups; women	women: 43 to 57%
Ethnicity	Afro-Colombians; Indigenous people	Afro share: 0 to 92%
Location	longitude and latitude; town center or rural; municipality; department	343 dots on the map

- Plus the **standing benchmark**: Petro's vote at the same station in 2018, where it existed.
- Before touching the x's, the classic first step: describe the y we want to predict (next slide). Session 3: the whole country; session 7: the 44 raw census columns.

The y, before any model

The classic first step: describe the target to know what getting it right means (training set only)



- The mass sits between 60 and 85 (77% of the stations), mean 69.1, standard deviation 12.2; the long tail runs downward: Petro loses at only 18 of the 257 (7%).
- Two lessons before fitting: predicting "the mean for everyone" already leaves a typical error of 12 points — that will be the **benchmark** — and the hard part will be in the tails (we return to this with the error by quintile).
- The census's candidate for starting to order this spread: **higher education**. Part 1.

1. Simple regression as a prediction machine
2. More flexible than the line, with one variable
3. Multiple regression
4. After predicting: understanding what the model predicts
5. Closing

The canonical chapter, read with fresh eyes

ISL opens chapter 3 with seven questions (about ads and sales); they are the same seven about the surroundings and the vote

1. Is there a relationship between a station's surroundings and its vote?
2. How strong is it?
3. Which variables of the surroundings are associated with the vote?
4. How large is each one's association?
5. **How accurately can we predict the vote of a station we have never seen?**
6. Is the relationship linear?
7. Are there interactions (does education weigh differently in city and countryside)?

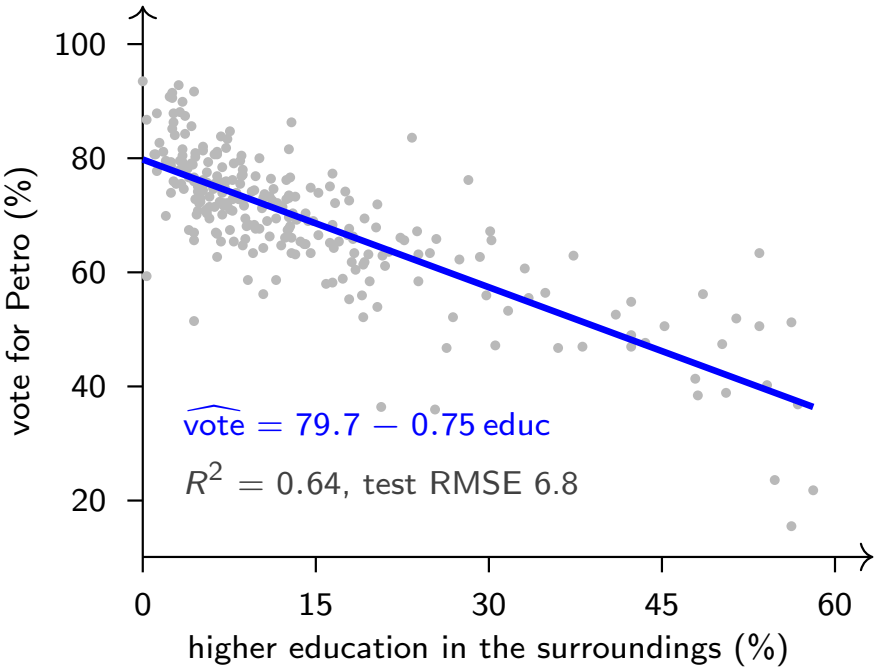
Six questions about the relationship and its coefficients; only the fifth is about predicting. And the chapter never evaluates out of sample.

Today: the itinerary

Question 5 at the center: the line and its mechanics, the metrics, more flexibility, more variables, and what comes *after* predicting. The rest fall along the way (or to the appendix).

The first answer: a line

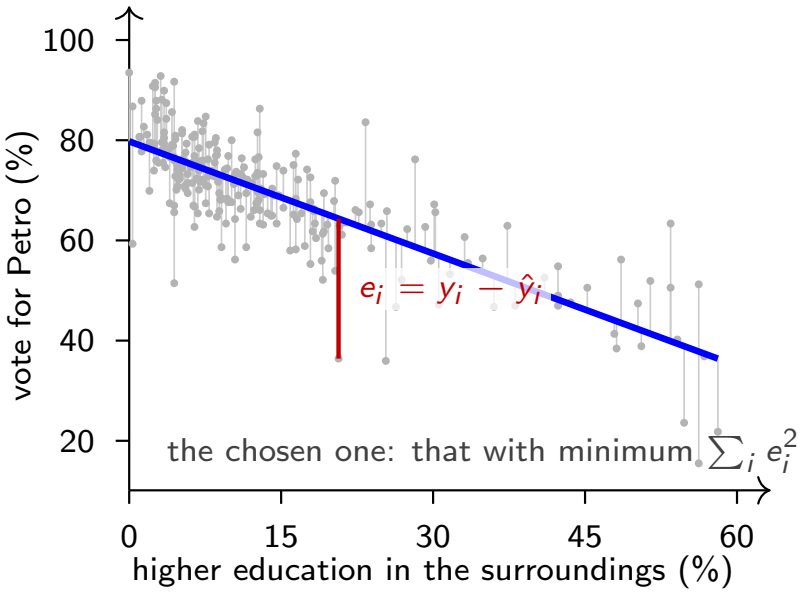
The 257 training stations against higher education, the census variable that moves most with the vote (cor -0.80)



- A single variable of the surroundings orders much of the map: $R^2 = 0.64$ in sample, 6.8 points of test RMSE (what they mean: three slides ahead).
- Drawing it is easy. The serious question: *which* line is this? Who chose it, by what criterion? Everything that follows lives off that answer.

Which line? The one that minimizes the RSS

Station i's residual: $e_i = y_i - \hat{y}_i$, what the line fails to explain about that station



The least-squares line minimizes

$$\text{RSS} = \sum_{i=1}^{n} e_i^2$$

- With the 257 stations: $\text{RSS} = 13.739$; $\text{RSS}/n = 53.5$, the **training MSE** (its root: 7.3, today's "in-sample RMSE").
- Minimizing the RSS *is* minimizing the training error (session 1's empirical risk).

Why the training error flatters (golden rule #1)

The line was chosen *in order to* make that error small: grading it with it is letting the student write their own grade.

The solution: $\hat{y}$ is the projection of y onto the x's

The same algebra from Econometrics, read in a prediction key

1. The problem: with $\mathbf{X}$ of size $n \times (p+1)$ (column of ones included),

$$\hat{\beta} = \arg\min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta).$$

2. First-order condition: $-2\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = 0 \;\Rightarrow\; \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$, and if $\mathbf{X}'\mathbf{X}$ is invertible,

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \qquad \hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \equiv \mathbf{H}\mathbf{y}.$$

3. The predictive reading: $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the space spanned by the columns of $\mathbf{X}$: the best *linear* approximation to the conditional mean. If $\mathbb{E}[Y \mid X]$ is linear, OLS estimates it without bias; if not, it estimates the best line, and the remaining bias is the price of rigidity.

4. Why collinearity does not scare the predictor: $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ depends on the *space* of the columns, not on how the coefficients split across similar columns. Two unstable coefficients can add up to a stable prediction. (The variance of $\hat{y}_0$ does grow with p: that is session 1's variance.)

The metrics: the test error has many faces

We know how to fit the line; what is missing is the number that grades it, always on the test set

Con n_t observaciones de prueba y predicciones $\hat{y}_i$:

Metric	Formula	When to use it
MSE / RMSE	$\frac{1}{n_t}\sum(y_i-\hat{y}_i)^2 \quad / \quad \sqrt{\text{MSE}}$	the standard; RMSE in units of y; punishes large errors
MAE	$\frac{1}{n_t}\sum\lvert y_i-\hat{y}_i\rvert$	robust to extremes; the "typical error"
MAPE	$\frac{100}{n_t}\sum\lvert y_i-\hat{y}_i\rvert/\lvert y_i\rvert$	relative error; useless when $y \approx 0$
Test R^2	$1-\sum(y_i-\hat{y}_i)^2/\sum(y_i-\bar{y}_{\text{train}})^2$	fraction of the mean's error the model avoids; **can be negative**

The metrics: the test error has many faces

We know how to fit the line; what is missing is the number that grades it, always on the test set

Con n_t observaciones de prueba y predicciones $\hat{y}_i$:

Metric	Formula	When to use it
MSE / RMSE	$\frac{1}{n_t}\sum(y_i-\hat{y}_i)^2 \quad / \quad \sqrt{\text{MSE}}$	the standard; RMSE in units of y; punishes large errors
MAE	$\frac{1}{n_t}\sum\lvert y_i-\hat{y}_i\rvert$	robust to extremes; the "typical error"
MAPE	$\frac{100}{n_t}\sum\lvert y_i-\hat{y}_i\rvert/\lvert y_i\rvert$	relative error; useless when $y \approx 0$
Test R^2	$1-\sum(y_i-\hat{y}_i)^2/\sum(y_i-\bar{y}_{\text{train}})^2$	fraction of the mean's error the model avoids; **can be negative**

- The RMSE in points says nothing alone: it reads **relative** to y. Today's line: 6.8 on a vote averaging 69 with spread 12: just over half the mean's error.
- RMSE $\geq$ MAE always; a large gap betrays **a few very large errors**. The line: 6.8 and 5.4, a short gap: its error is even.
- The test R^2 **can be negative**: worse than predicting $\bar{y}$ (appendix A7). Always report **more than one**.

1 Simple regression as a prediction machine

2 More flexible than the line, with one variable

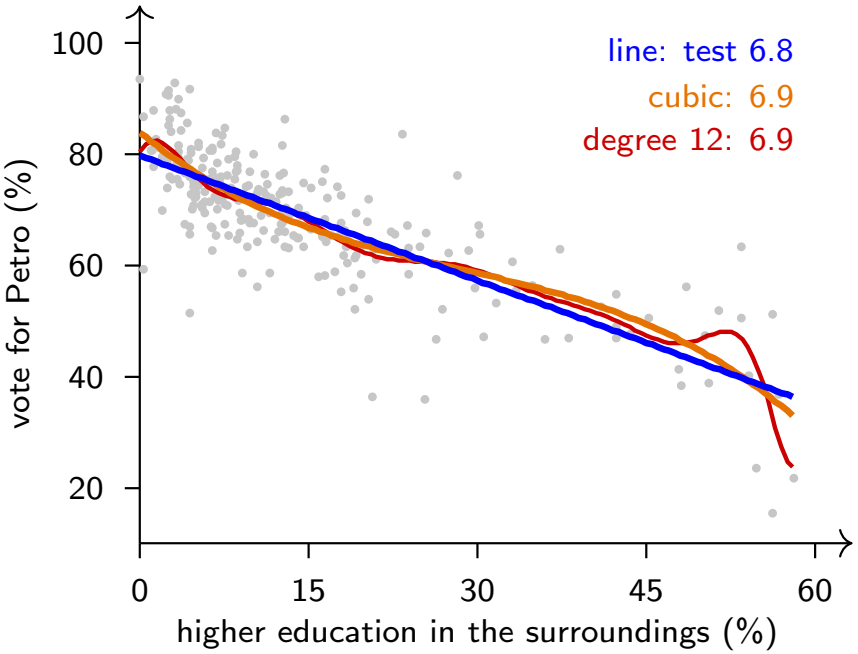
3 Multiple regression

4 After predicting: understanding what the model predicts

5 Closing

Polynomials, with our stations: they hug each other

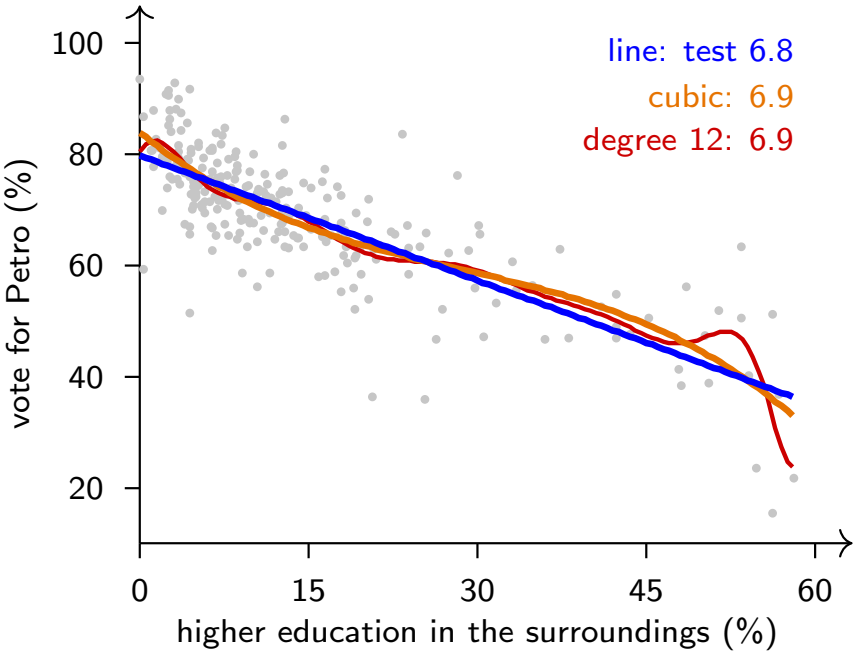
Within the observed range, the line, the cubic, and degree 12 are almost the same function



- A degree-d polynomial is still OLS on the columns $x, x^2, \ldots, x^d$; the degree is the knob. Here the relationship is nearly linear and the flexibility **bought nothing** (test: 6.8 / 6.9 / 6.9); in ISL's `Auto` (Fig. 3.8) the curvature is real and the quadratic does win.
- Degree 12 already *wiggles* where the data thin out: it pays variance without removing bias. And outside the range the three truly part ways (extrapolation, at the close).

Polynomials, with our stations: they hug each other

Within the observed range, the line, the cubic, and degree 12 are almost the same function



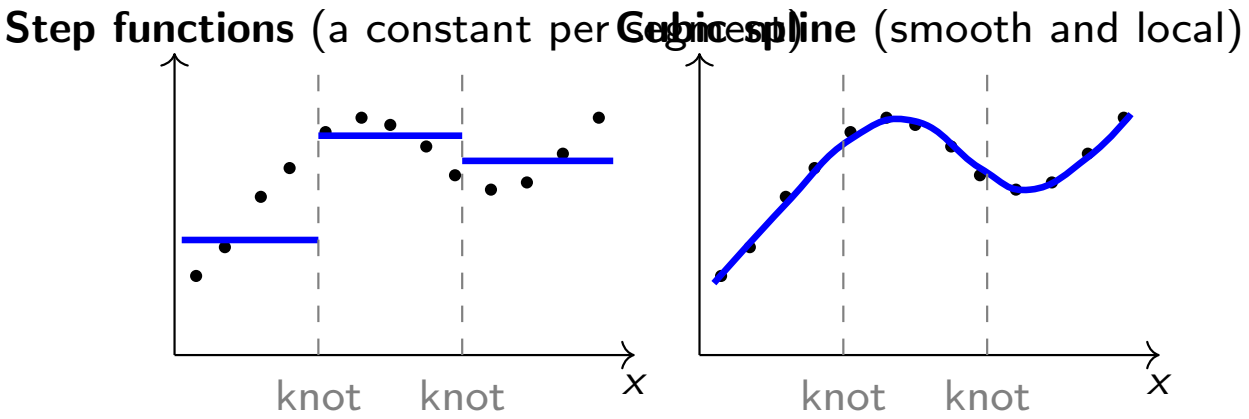
- A degree-d polynomial is still OLS on the columns $x, x^2, \ldots, x^d$; the degree is the knob. Here the relationship is nearly linear and the flexibility **bought nothing** (test: 6.8 / 6.9 / 6.9); in ISL's `Auto` (Fig. 3.8) the curvature is real and the quadratic does win.
- Degree 12 already *wiggles* where the data thin out: it pays variance without removing bias. And outside the range the three truly part ways (extrapolation, at the close).

What we want

Local flexibility: the shape of the relationship in one zone of x should force nothing in another. That is what step functions and splines deliver.

Beyond linearity, in one slide

Step functions, splines, and additive models: the ladder that is still OLS (ISL, ch. 7)



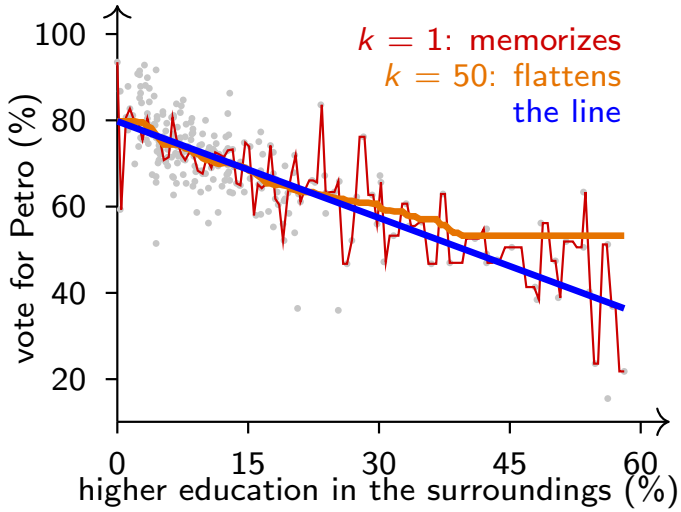
- **Step functions:** cut x into segments and fit a constant per segment (range dummies). Easy to read, ugly at the edges.
- **Cubic spline:** a degree-3 polynomial per segment, forced to join smoothly at the **knots**. Still OLS on a *basis* of columns built from x (appendix A5); the knob is the number of knots; *natural splines* are linear in the tails.
- **GAM:** $y = \beta_0 + f_1(x_1) + \cdots + f_p(x_p) + \varepsilon$, one spline per variable, added up. Flexible where it matters and readable variable by variable (that figure returns in session 6 as partial dependence); its limit is that it discovers no interactions.

Dropping the formula: k-NN, the neighbors' average

All the previous knobs bend a global formula; the other extreme is having none

To predict at x_0: average the y of the k **nearest** observations:

$$\hat{f}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} y_i$$



- **No formula**: the data supply the shape. The knob is k: at $k = 1$ the curve passes through every point (variance); at large k it flattens toward the mean (bias).
- It needs a **distance** (standardize the x's). Here the neighbors live along *education*; today's final table also tries the **map**. In session 4 it votes; today it averages.

The curse of dimensionality

Why "the neighbors" stop being neighbors

With predictors uniform on $[0,1]^p$, what fraction of *each* variable's range must you span to capture 10% of the data as neighbors?

p	1	2	5	10	100
Side of the "neighborhood" $(0.1^{1/p})$	0.10	0.32	0.63	0.79	0.98

- With $p = 10$, the "neighborhood" that gathers 10% of the data covers 79% of each variable's range: the neighbors no longer resemble x_0 in anything. The local average becomes a *global* one and the bias explodes.
- Methods that rely on distances (k-NN, kernels) take this head on; linear models, additive splines, and trees dodge it because they **impose structure**.

Lesson

Total flexibility is only free with few variables and lots of data. In high dimension, some *structure* (linearity, additivity, partitions) is what makes learning possible. That is the underlying argument of sessions 3 and 5.

1 Simple regression as a prediction machine

2 More flexible than the line, with one variable

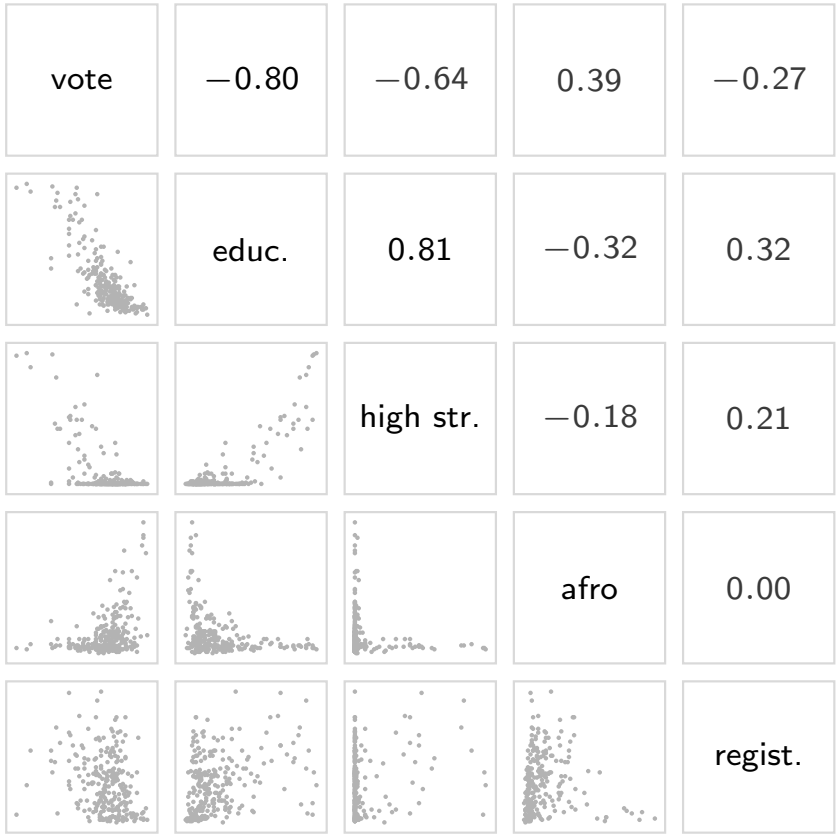
3 Multiple regression

4 After predicting: understanding what the model predicts

5 Closing

Look before you fit: everything against everything

Five of the variables, with the 257 training stations only: exploration also respects the split

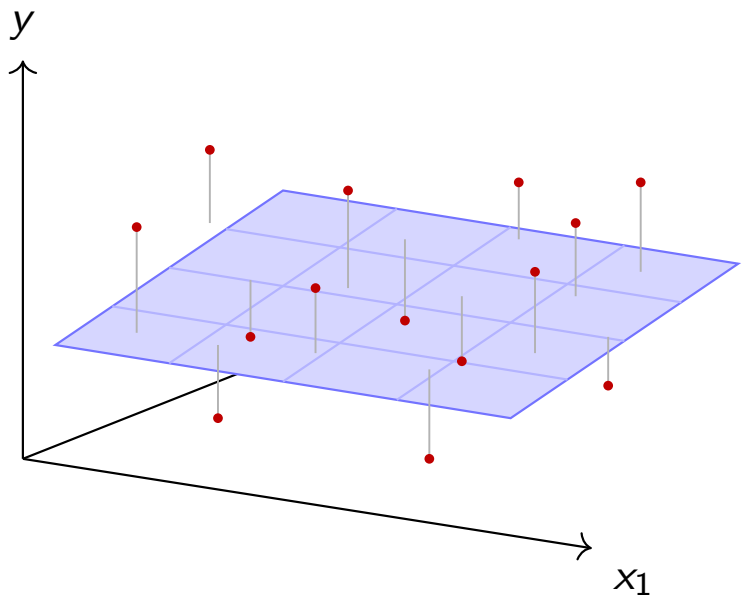


- **The vote's row** ranks the candidates: higher education (−0.80) is already in the model; high stratum, Afro share, and registered voters await their turn.
- **Twins in sight:** education and high stratum (0.81) tell almost the same story. A few slides ahead: what happens if they enter together.
- **Registered voters** is skewed (median 6,100, maximum 22,000): when it enters the model, it will enter in logs.

With 9 variables you can still look at everything; with session 3's hundreds, no longer. And watch the **twins** (0.81): that bill comes due in three slides.

From the line to the plane, and beyond drawing

The same criterion with two predictors: minimize the squared vertical distances to the plane



- With $p = 2$ it can still be drawn: the RSS adds vertical distances to a **plane**. With our nine variables, the fit is a hyperplane in a space nobody can see.
- The practical consequence: from there on **evaluation is numerical** and the intuition of "near" must be earned (the previous part's curse). And does the in-sample report notice that variables came in? Next slide.

The in-sample report: RSE and R^2

The two measures you already knew, reread; and a warning when adding variables

- **RSE** $= \sqrt{\mathrm{RSS}/(n-2)} = 7.34$ points: the residual's standard deviation. Reread for prediction: the **estimate of** σ, session 1's ceiling. Keep it: it sets the width of the prediction band ($\approx 4 \times \mathrm{RSE} \approx 29$ points; appendix A1).
- $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS} = 0.64$, **in sample**: the fraction of the vote's variance the line orders *in the very data that chose it*.
- When variables are added, the in-sample R^2 **never falls** — even if the variable is garbage — and the RSE barely flinches. How to know, then, whether the new variable is expendable? Question for the room, next slide.

The rule

In-sample fit is not predictive ability. Every "official" number in this course is reported on the **test set**; this slide's numbers describe the fit, they do not grade it.

Question for the room: what if I add a twin?

As promised in the pairs matrix: the R^2 does not even notice; the damage is elsewhere

Model	R^2	Adj. R^2	se(education)	Test RMSE
Education alone	0.640	0.638	0.035	6.8
+ high stratum (cor 0.81)	0.640	0.637	0.060	6.8
+ a twin of education (cor 0.999)	0.640	0.637	0.911	6.8
+ 150 columns of pure noise	**0.853**	0.642	0.056	**9.9**

- The in-sample R^2 **never falls** when a variable is added: that is why it cannot choose them. With pure noise it reaches 0.85; the adjusted one discounts it and the test set charges for it (9.9 vs. 6.8).
- With the twin the predicted point barely changes, but the **credit is split at random**: education's -0.75 splits into -0.24 and -0.51, with a standard error 26 times larger. Interpreting coefficients there is reading noise (why the prediction walks away unharmed: part 1's projection).
- And the hidden risk: the cancellation between twins works *as long as their correlation holds*; if they decouple in new data, the prediction breaks ("the line does not travel").

OLS with fresh eyes

The twin's lesson, in general: $\hat{\beta}$ and $\hat{y}$ answer different questions

The model and the estimator are those of Econometrics I:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \;=\; \arg\min_{\beta} \mathrm{RSS}(\beta)$$

What changes is **what for**:

	As econometricians	**As predictors**
We care about	$\hat{\beta}_j$ and its standard error	$\hat{y}_0 = x_0'\hat{\beta}$ on new data
We choose variables	by theory and identification	by test error
An "insignificant" variable	gets debated	stays if it helps predict
Unbiasedness	what gets defended	tradable for less variance
Collinearity	a problem for $\hat{\beta}$	nearly irrelevant for $\hat{y}$
The model's test	assumptions, R^2, tests	the error on the test set

OLS's role in the course

OLS approximates the conditional mean, is stable, fits in milliseconds, and is read effortlessly. Every method in the course is compared against it: if a forest does not beat it on the test set, the forest is expendable.

1 Simple regression as a prediction machine

2 More flexible than the line, with one variable

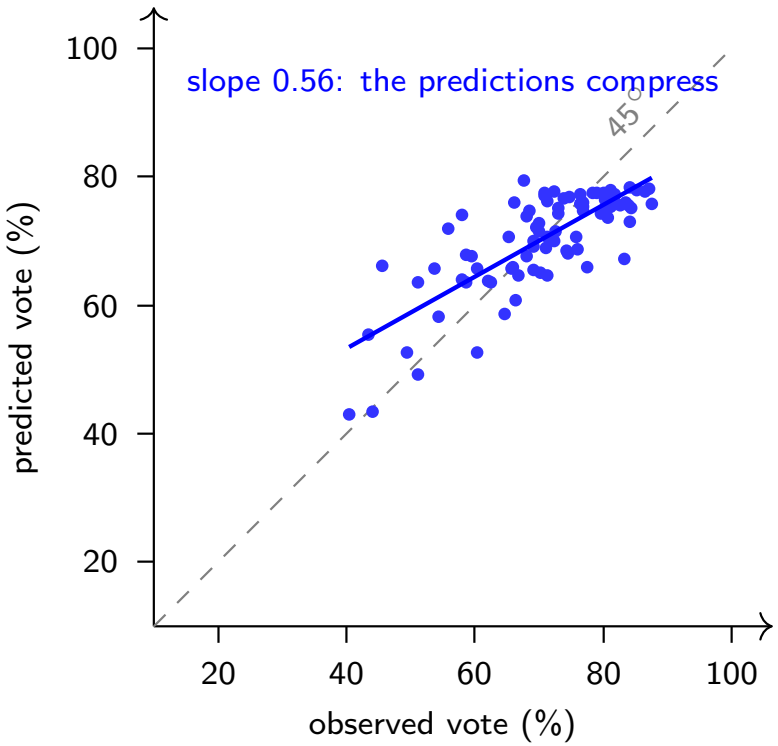
3 Multiple regression

4 After predicting: understanding what the model predicts

5 Closing

Predicted vs. observed: the figure that never goes missing

What the model predicts: the line, on the 86 test stations



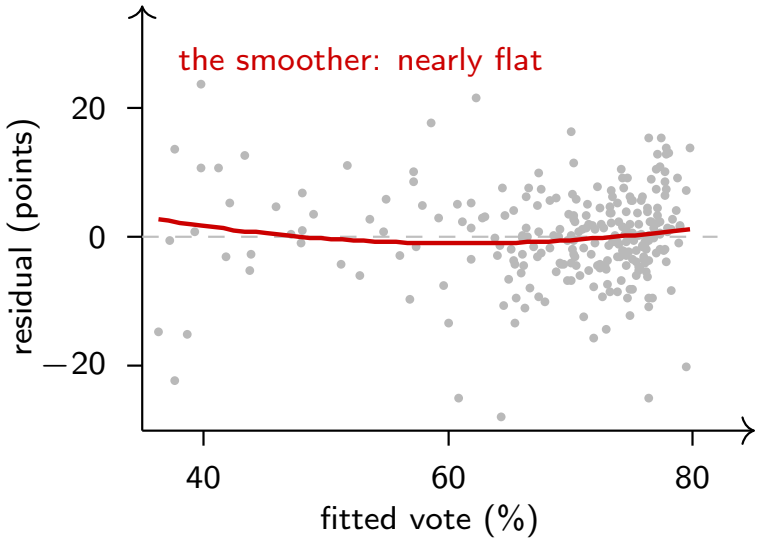
- Each dot is a **test** observation; the diagonal is the perfect prediction.
- The typical pattern: **slope below one** (here, 0.56). The low tail is **overpredicted** and the high tail **underpredicted**: the predictions *compress* toward the mean (the algebra: appendix A8).
- The report's first figure: one per model, on the same test set.

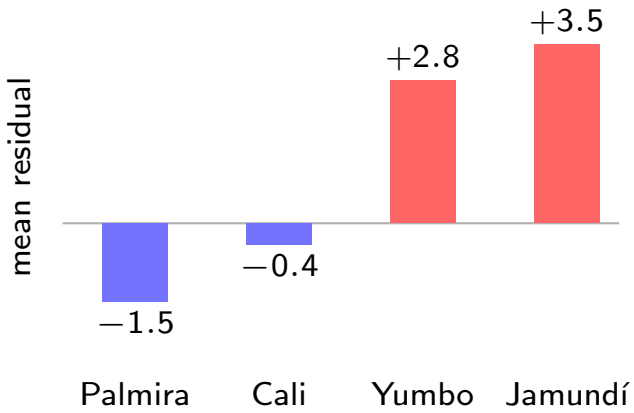
What to look for in the figure

Curvature (omitted nonlinearity), fanning (variance growing with y), shifted groups, and isolated points (large errors). The R^2 summarizes; the figure explains.

Residuals as a detector

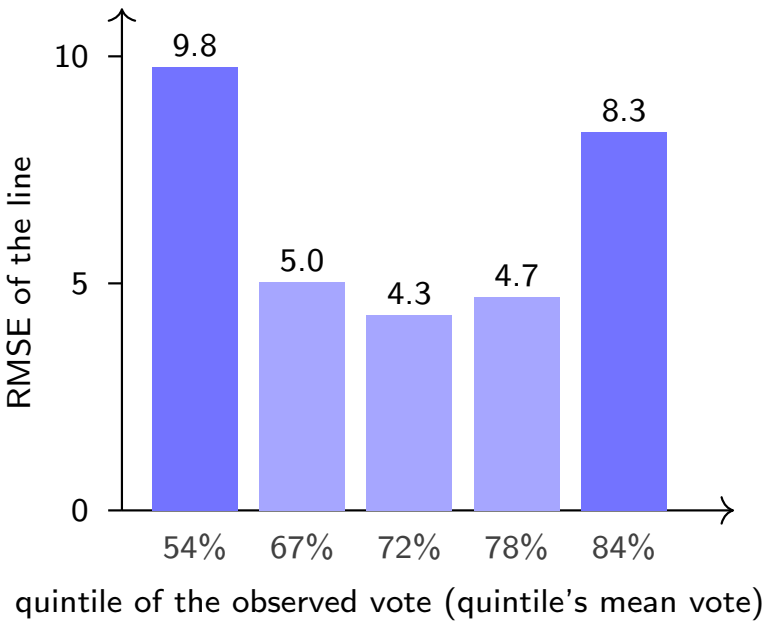
Plot e_i against whatever you suspect: wherever there is a pattern, structure was left uncaptured





- Against the **fitted values** (left), nothing: this line is not missing curvature (hence the cubic bought nothing). In ISL's `Auto` this panel shows a U: the quadratic was missing.
- Against the **municipality** (right), yes: short in Jamundí and Yumbo, over in Palmira. It is not missing shape: it is missing *place*. The fine version is the map (two slides).
- The general detector: if anything (a group, the map, time, another method) can *predict* your residuals, your model left structure out.

Where the error concentrates (I): by quantiles of y



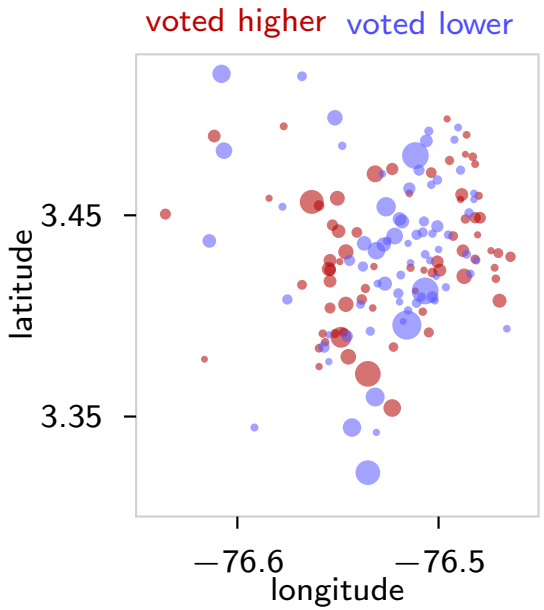
- Compute the RMSE **by decile** (or quintile) of y on the test set. The U is the compression's footprint: the extremes fail most.
- Today: 9.8 at the bottom (54% vote), 5.0, 4.3, 4.7 in the middle, 8.3 at the top (84%): the near-unanimous, *exactly* what a campaign wants to spot.
- Automated valuation models report R^2 by price quartile for the same reason: the aggregate hides the tails.

Two remedies we will see

More flexibility where the relationship bends (session 5); sometimes, another loss: if the high tail costs more, symmetric MSE is not the objective.

Where the error concentrates (II): subgroups and the map

The same question, cut by **who** and by **where**



- **On the map:** the line's residuals at Cali's 143 stations (dot size is the error). Same-sign patches betray omitted *location* variables: that is why validation with spatial data demands care (session 3).
- **By subgroup:** RMSE by municipality, by zone (town center or rural), by stratum. A good overall RMSE that fails for one group is a **fairness** problem as well as an accuracy one (sessions 4 and 6).

The habit worth acquiring

After the metrics table, the figures: predicted vs. observed, residuals against whatever you suspect, and the error by decile of y and by the subgroup (or map) the question cares about.

Always against a benchmark

A metric alone does not say whether the model is good. Three comparisons that never go missing:

1. **The mean** (or the median): the model that uses no x. If you cannot beat it, there is no learnable signal. Today: predicting 69% for everyone gives an RMSE of 11.0; the line, 6.8.

2. **The simple model**: OLS with the obvious variables; everything else must justify its complexity against it. Today's line is that model.

3. **The standing rule**: the cadastral appraisal, the Sisbén, the analyst's forecast. For a campaign, *what happened last time*: the 2018 vote at the same station. As is, 14.1 (Petro grew 11 points); with a line that corrects the level, 8.4; the education line on those 319 stations, 7.2. You must predict better than what is already in use.

How it is reported

A table with one row per model and one column per metric, all computed on the *same* test set, with the benchmark in the first row. Without that row, "RMSE 6.8" cannot be read. Ours: next slide. And it is inherited every week.

Today's canonical table

All of part 1 in one table: eight models, the same test set (86 stations), three metrics, and the benchmark

Model	In-sample RMSE	Test RMSE	Test MAE	Test R^2
M0: training mean (69.1)	12.2	11.0	8.9	0.00
M1: line (higher education)	7.3	6.8	5.4	0.62
M3: cubic	7.1	6.9	5.5	0.61
M12: degree 12	7.0	6.9	5.5	0.61
M9: nine variables	6.8	6.8	5.1	0.63
k-NN on the map, $k = 1$	0.0	8.9	6.8	0.35
k-NN on the map, $k = 5$	6.4	8.2	6.2	0.44
k-NN on the map, $k = 25$	9.7	8.6	6.8	0.39

- The first row is the benchmark: without it, "6.8" says nothing. The line avoids 62% of the mean's error; eight extra variables, the cubic, and degree 12 add nothing on the test set. **At equal error, the simplest: today's champion is the line.**
- $k = 1$: the smallest error **in sample**, the worst **out of sample** (golden rule #1). And the neighbors lose to the line: education is worth more than location.
- **The standing rule** (the 2018 vote at the same station, corrected with a line): 8.4. The census beats the past.

1 Simple regression as a prediction machine

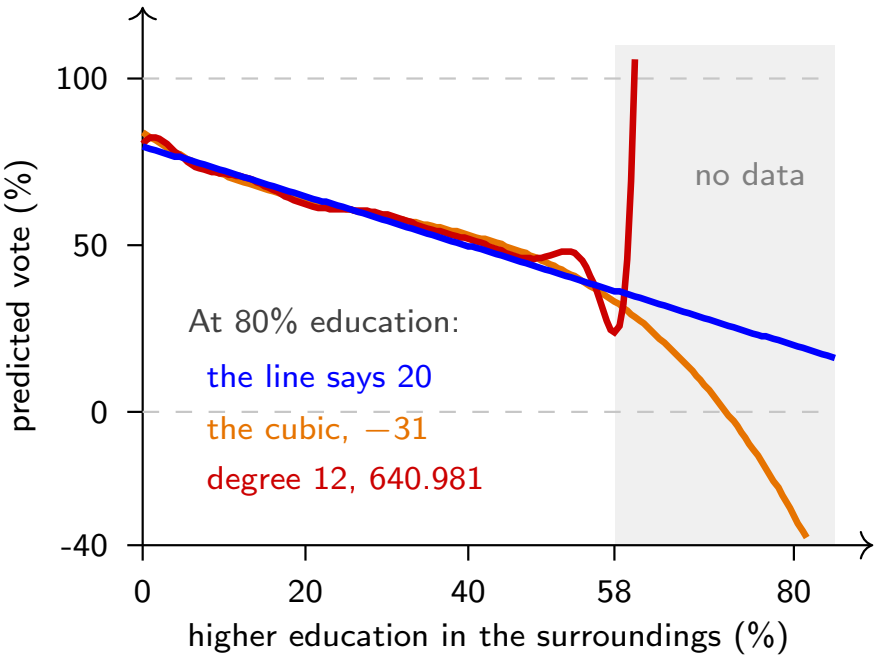
2 More flexible than the line, with one variable

3 Multiple regression

4 After predicting: understanding what the model predicts

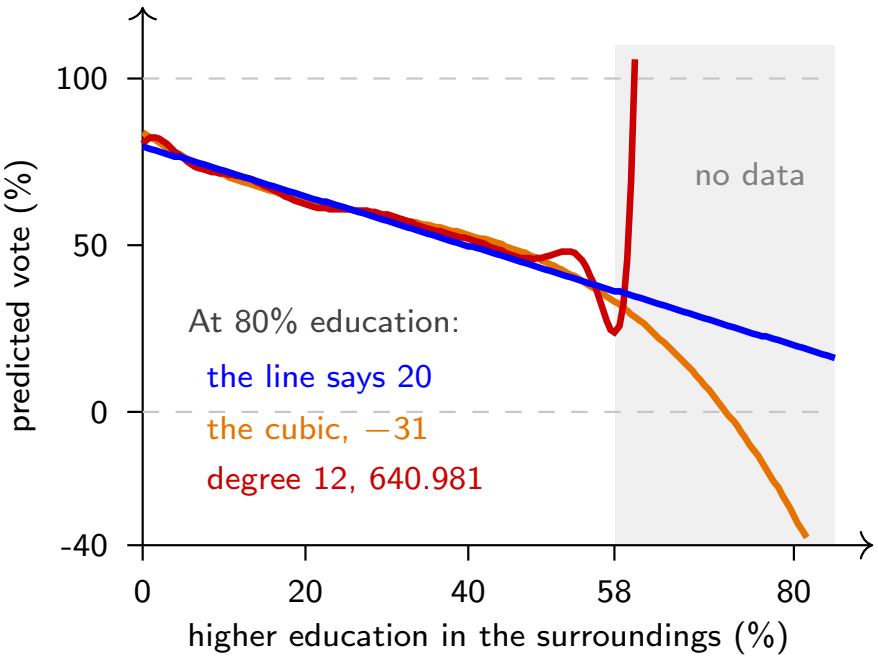
5 Closing

Extrapolation: predicting where there are no data



- Outside the training range no model *knows*: the polynomial shoots off; k-NN and trees stay **flat** at the last value.
- The line extrapolates "reasonably" only if the relationship stays linear out there, which the data cannot confirm. *Natural splines* impose linearity in the tails: an explicit compromise.

Extrapolation: predicting where there are no data



- Outside the training range no model *knows*: the polynomial shoots off; k-NN and trees stay **flat** at the last value.
- The line extrapolates "reasonably" only if the relationship stays linear out there, which the data cannot confirm. *Natural splines* impose linearity in the tails: an explicit compromise.

Rule

Report the predictors' **range** in the training set and flag predictions outside it: a model trained on surroundings with 0 to 58% higher education knows nothing about one with 80% (today: the line says 20, the cubic −31, and degree 12, 640,981).

The acid test: the line does not travel

So far the line played at home; the same line, applied to the rest of Valle del Cauca:

Stations	n	Actual vote	Predicted vote	Line RMSE	S.d. of y
Metro area (test)	86	70.7	70.3	6.8	11.0
Rest of Valle, town centers	169	58.3	71.5	21.2	16.1
Rest of Valle, rural	465	52.4	77.9	33.9	21.8
Buenaventura	83	91.4	75.4	17.6	8.1

- Outside the four cities the line predicts **worse than the local mean**: in the countryside it says 78 where reality is 52; in Buenaventura, 75 where it is 91. The nine variables improve little (31.8 and 11.1).
- And it is not extrapolation (education there falls *within* the range): the *relationship* changes. Low education in Cali is a working-class barrio that votes Petro; in northern Valle, a coffee hamlet that does not. Country-wide, the education line explains **0.04%**: Santander votes 20% and Chocó 86% with the same education.
- **With Cali's housing:** the same canonical table is Part 1 of Problem Set 1.

The limitation that takes us to session 3

We need the whole country and many more variables (regions, interactions, all the blocks). With hundreds of columns, how do we choose without fooling ourselves, and how do we validate when neighboring stations look alike?

Recap

1. **The line as a machine:** the RSS it minimizes is the *training* error (that is why it flatters); $\hat{y}$ is the **projection** of y onto the x's; the grade is set on the **test set**: RMSE, MAE, test R^2, all relative to y.
2. **More flexible with one variable:** the polynomials hug each other here; step functions and splines give local flexibility; k-NN drops the formula and the **curse** collects: without structure there is no learning in high dimension.
3. **More variables:** the matrix shows candidates and twins; the criterion is the same RSS; the in-sample R^2 never falls and so cannot choose variables; twins split credit, not prediction (appendix A9).
4. **After predicting:** predicted vs. observed, residuals as a detector, the error by quintile and by subgroup or map, all against a **benchmark** and summarized in the **canonical table**. The message: understand *what* it is the model predicts.
5. **The two warnings:** extrapolating (predicting where there are no data) and traveling (predicting where the relationship changes); the second opens session 3.

Session 8: one paper per field of economics. Pick yours today

November 3: six 15-minute presentations, six different fields

Field	**The typical question**
Political economy	How much of the vote (or of conflict) can territorial data anticipate?
Crime and justice	Whom to release before trial: the judge or the algorithm?
Health	Where does the doctor err when assigning tests and care?
Development and poverty	How to target aid where there is no census (phones, satellites)?
Labor	Hire by the standardized test or by the recruiter's nose?
Finance and credit	Who wins and who loses when a machine decides credit?
Macroeconomics	Does a machine with hundreds of predictors forecast inflation better?
Industrial organization	How much is personalized pricing worth, and who pays for it?

Today you pick the field; each field's two candidate papers, with their summaries, are in `literature/`, and the assignment closes this week. **The 15-minute script:** the question and the decision; the data and the unit; the benchmark; the out-of-sample evaluation and where it fails; the result; which method of the course it is.

Getting ready for next class...

To get the most out of next session

This week's readings:

- ISL, section 5.1 (cross-validation) and sections 6.2 and 6.4 (Ridge, Lasso, and high dimension).
  Guide: figure 5.6 (the CV curve) and figures 6.4 and 6.6 (the Ridge and Lasso paths) are the ones from class.

R leveling: continue with the data input/output and data management modules (links in session 1).

Next session — cross-validation and regularization: choosing between models without spending the test set (k-fold, the CV curve, the leak-free pipeline, validating by groups, over time, and across space) and what to do with many predictors (Ridge, Lasso, and Elastic Net, with λ by CV). **R application:** the country's 12,000 stations and the 37 census variables: cross-validation fold by fold, Ridge and Lasso with λ by CV, and the scale (standardize before penalizing).

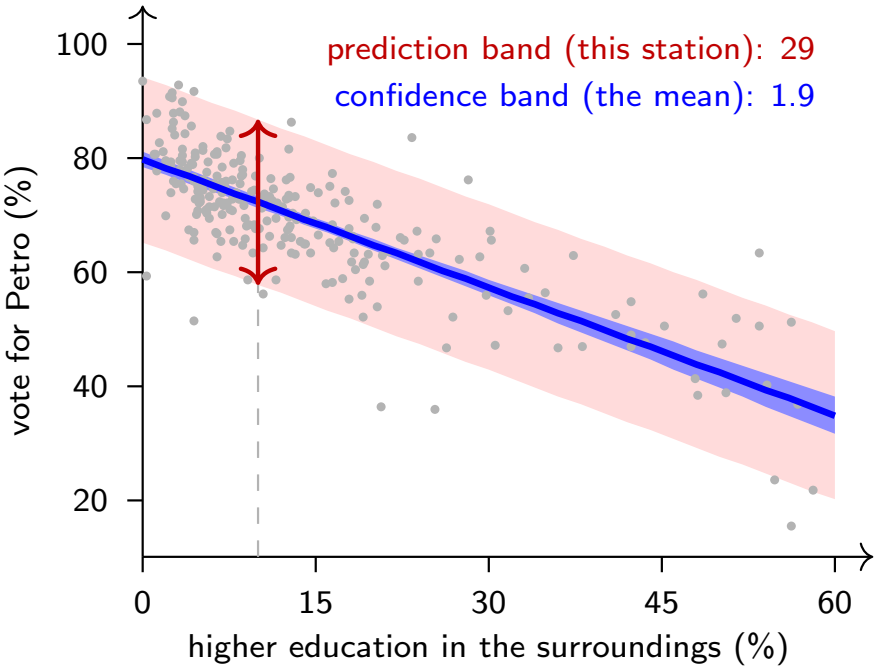
Essential references

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2. ed.). Springer. Chs. 2–3 and 7. [ISL]
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2. ed.). Springer. Chs. 2–3 and 5. [ESL]
- Mullainathan, S., & Spiess, J. (2017). Machine Learning: An Applied Econometric Approach. *JEP*, 31(2), 87–106.
- Gelvez, J. D., Cardiles, A., Martínez-González, E. F., & Muñoz, M. (2026). How Predictable Is an Election? A Machine-Learning Approach to Electoral Behavior in Colombia. Working paper. [The source of the case data.]
- Session 8's candidate papers are in `literature/`, with their summaries, organized by field of economics.

6 Appendix: going deeper

A1. Predicting the mean or predicting a new data point

From $\hat{\beta}$ to $\hat{y}_0$: two intervals, and only one narrows with more data (ISL, pp. 82–83)



Three errors when predicting $y_0 = f(x_0) + \varepsilon_0$**:**

1. estimating $\hat{\beta}$ from a sample: reducible, shrinks with n;
2. assuming f is linear: the model's bias;
3. ε_0: irreducible, nobody predicts it.

The **confidence interval** covers the mean $f(x_0)$: only the first error. The **prediction interval** covers *this* station's vote and adds σ^2: always wider.

With the 257 training stations, in surroundings with 10% higher education

Confidence interval width: 1.9 points; prediction interval: 29.0 ($\approx 4 \times \mathrm{RSE}$). With 65 stations: 3.6 and 27.9. The blue narrows with data; the red does not: session 1's ceiling, the interval that matters to the user.

A2. Qualitative predictors

So far the x's were numbers; categories enter as columns of zeros and ones

A variable with K categories enters as $K - 1$ dummies: the zone (town center or rural), the municipality, the department. Two readings for the predictor:

- The coding (which category is the base) changes the coefficients and their p-values, **not** the predictions.
- What does matter: do not leave categories with two or three observations — those coefficients are noise and travel badly.

In the case

Today's line deliberately uses neither zone nor municipality: they are exactly what the residual detector interrogated, and what session 3 puts into the model when we move to the whole country.

A3. Interactions: the effect depends on where

El efecto de una variable puede depender de otra:

$$\text{voto} = \beta_0 + \beta_1\,\text{educ} + \beta_2\,\text{rural} + \beta_3\,(\text{educ} \times \text{rural}) + \varepsilon$$

Higher education moves the vote differently in town centers and in the countryside (and, we will see, in Cali and on the Pacific coast): without the interaction, the line errs *systematically* in both.

- Hierarchy rule (ISL): if the interaction enters, both main effects enter, significant or not.
- They are **flexibility knobs**: each adds parameters and lowers bias at the cost of variance. With 17 variables and 33 departments they are hundreds of columns: picking them by hand does not scale. The Lasso (session 3) filters them; trees (session 5) discover them on their own.
- **Practical rule:** add the ones economics suggests (education $\times$ region, age $\times$ zone) and let the **test set** say whether they were worth it.

A4. Transformations: the logarithm and polynomials

Log of y: with prices, incomes, or sales, modeling $\log y$ stabilizes the variance and turns multiplicative effects into additive ones. Careful on the way back: $\exp(\widehat{\log y})$ predicts the *median*, not the mean; the error is reported on the scale the user cares about (pesos, not logs). Our y is a proportion: the line can predict outside 0–100 (we will see it when extrapolating).

Polynomials: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d + \varepsilon$ is still **linear in the parameters**: it is OLS on new columns.

- The degree d is session 1's complexity knob: more degree, less bias and more variance.
- A polynomial is **global**: to bend the relationship in low-education surroundings, it also changes its shape in high-education ones. And outside the data's range, it explodes.

A5. A spline's basis: how the columns are built

Piecewise cubic with one knot at ξ: two different cubics on either side of ξ, joined so the function, its first, and its second derivative are continuous at ξ.

The truncated power basis achieves it with a single extra column:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3 + \varepsilon, \qquad (x - \xi)_+^3 = \max\{0, (x - \xi)^3\}.$$

Before ξ the column equals zero; afterwards it adds a cubic that starts at zero with zero derivatives: the joint is smooth by construction.

- With K knots: $K + 4$ columns, that is, $K + 4$ degrees of freedom. Everything is estimated by OLS.
- A **natural spline** imposes linearity beyond the outer knots and saves 4 degrees of freedom: less variance in the tails, where data are few.
- In R, `splines::ns(x, df = 4)` builds the basis and is used inside `lm()`. *B-splines* are another, numerically more stable, basis of the same space.

A6. Linear vs. k-NN: when does each one win?

ISL, section 3.5: the chapter's only one with the test error on the axes

- If f is (nearly) linear, **OLS wins**: it estimates two numbers; k-NN spends the data drawing a shape it did not need. If f is nonlinear and p is small, **k-NN can win**.
- As p grows, **OLS wins again even when misspecified** (ISL, Fig. 3.20: k-NN wins with $p \leq 2$; the line, from 4 on). Few observations per predictor favor the parametric.
- **Today:** the line with one good variable, 6.8; the map's neighbors, 8.2 (part 4's canonical table). With 257 stations, education is worth more than location; in session 3, with the country, the arithmetic changes.

A6. Linear vs. k-NN: when does each one win?

ISL, section 3.5: the chapter's only one with the test error on the axes

- If f is (nearly) linear, **OLS wins**: it estimates two numbers; k-NN spends the data drawing a shape it did not need. If f is nonlinear and p is small, **k-NN can win**.
- As p grows, **OLS wins again even when misspecified** (ISL, Fig. 3.20: k-NN wins with $p \leq 2$; the line, from 4 on). Few observations per predictor favor the parametric.
- **Today:** the line with one good variable, 6.8; the map's neighbors, 8.2 (part 4's canonical table). With 257 stations, education is worth more than location; in session 3, with the country, the arithmetic changes.

The course's comparison rule

You never know a priori. Fit both on the same training set, evaluate them on the *same* test set with the *same* metric, and the winner wins. That comparison table is the central product of every application in the course. The moral: with 257 stations and few variables, **structure is information**. The k is chosen without touching the test set: session 3.

A7. Test R^2 is not Econometrics' R^2

Part 1's in-sample R^2, reincarnated where it cannot cheat

The R^2 you know (in sample):

- It never falls when variables are added.
- It sits between 0 and 1 by construction.
- It measures fit, not predictive ability: degree 12 had the highest R^2 of the three.

The test R^2:

- **It can fall** when variables are added: it is the U curve.
- **It can be negative**: the model predicts worse than the training mean.
- It compares against an explicit benchmark: predicting $\bar{y}_{\text{train}}$ for everyone.

A7. Test R^2 is not Econometrics' R^2

Part 1's in-sample R^2, reincarnated where it cannot cheat

The R^2 you know (in sample):

- It never falls when variables are added.
- It sits between 0 and 1 by construction.
- It measures fit, not predictive ability: degree 12 had the highest R^2 of the three.

The test R^2:

- **It can fall** when variables are added: it is the U curve.
- **It can be negative**: the model predicts worse than the training mean.
- It compares against an explicit benchmark: predicting $\bar{y}_{\text{train}}$ for everyone.

How to read it, with the stations

The line: test $R^2 = 0.62$: on stations it never saw, its squared error is 38% of the average-predictor's; $k = 1$ avoids only 35%. It can be negative: session 1's degree 12 scored -0.90 on the new listings. An *error reduction*, not a "goodness of fit".

Detail for the careful: the denominator uses the **training** $\bar{y}$; the metrics are computed **once** on the test set.

A8. The compression algebra: $\mathrm{Var}(Y) = \mathrm{Var}(f(X)) + \sigma^2$

1. Law of total variance: for any Y and X,

$$\mathrm{Var}(Y) = \mathbb{E}\big[\mathrm{Var}(Y \mid X)\big] + \mathrm{Var}\big(\mathbb{E}[Y \mid X]\big).$$

2. With $Y = f(X) + \varepsilon$, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathrm{Var}(\varepsilon \mid X) = \sigma^2$: $\mathbb{E}[Y \mid X] = f(X)$ and $\mathrm{Var}(Y \mid X) = \sigma^2$, hence

$$\mathrm{Var}(Y) = \sigma^2 + \mathrm{Var}\big(f(X)\big) \;\geq\; \mathrm{Var}\big(f(X)\big).$$

3. Calibration and slope: if $\hat{y}$ is well calibrated, $\mathbb{E}[Y \mid \hat{y}] = \hat{y}$, so in the regression of y on $\hat{y}$ the slope is 1 and the intercept 0. The *reverse* regression, of $\hat{y}$ on y, has slope $\mathrm{Cov}(y, \hat{y})/\mathrm{Var}(y) = \mathrm{Var}(\hat{y})/\mathrm{Var}(y) \leq 1$: that is the compression you see when plotting $\hat{y}$ against y.

Use

The slope of y on $\hat{y}$ on the test set is a free calibration test: below 1 there is overfitting (predictions too spread out); above, underfitting. It returns in session 4 for probabilities.

A9. The six problems, reread for prediction

ISL, section 3.3.3: what they damage in $\hat{\beta}$ and what they damage in $\hat{y}$

Problem	**What it damages in $\hat{\beta}$**	**And for $\hat{y}$?**
Nonlinearity	everything the coefficients say	**Yes**, the only one ISL names: new variables or flexible methods
Correlated errors	understated standard errors	The point does not change; **validation** is damaged: stations in the same municipality look alike (session 3)
Non-constant variance	standard errors and tests	The prediction intervals, not the point
Outliers	the RSE and everything that depends on it	The line barely moves; they damage the **metrics** and signal a missing predictor
High leverage	a couple of points move the fit	**Yes**: every prediction moves ("one new data point changes the curve")
Collinearity	huge standard errors, sinking t's	**Almost nothing**: dropping the collinear variable leaves the fit intact (ISL, p. 103); Ridge turns it into an advantage (session 3)

In high dimension (ISL, section 6.4.4)

You never know which variables are "the true ones": another subset predicts just as well. The model is useful; reading it as if it were unique is not.